4.1 Cardiac Arrest

Treat cardiac arrest patients in a systematic prioritized way ensuring immediate and sustained high-quality BLS. High-quality chest compressions should then be integrated and coordinated with Advanced Live Support (ALS) procedures. Reversible causes and special circumstances should be considered early in the resuscitation and treated accordingly.

4.1.1 Basic Life Support

Chest Compressions

Establish pulselessness and immediately initiate BLS ensuring high quality compressions are performed at all times

- Compression rate at 100–120 compressions per minute
- Compression depth 5–6 cm in adults or 1/3 Anterior Posterior diameter in children and infants
- Complete chest recoil after each compression
- Avoid excessive ventilation
- Minimize interruptions to chest compressions
- Rotate compressor every 2 minutes or use a mechanical compression device

  NOTE: **Mechanical chest compression devices should be used when they are available**
- Continuous compressions with passive oxygen insufflation using the i gel or basic airway adjuncts may be used as the initial BLS approach for up to 3 cycles or 200 continuous compressions

Ventilation and Oxygenation

Ensure cardiac arrest specific ventilation is performed at all times

- Avoid excessive ventilation and continually monitor ventilation rate
- Ventilation pressure and volume should be “just enough” to make the chest rise
- Administer O2 at the highest FiO2 available

Mask Ventilation

If a bag and mask is used to ventilate during CPR:

- Insert an OPA, NPA, or both in combination
- Remove the PEEP valve or set it to zero
- Use a 2 handed jaw thrust / thumbs down mask grip/2 person technique
- A PHI Medical Crewmember should manage the airway and mask
- Pause compressions during mask ventilation

Compression Ventilation Ratio

Maintain a 2 rescuer compression to ventilation ratio of:

- Adult—30:2
- Child/Infant—15:2

Advanced Airway Ventilation

If an advanced airway (ETT, King, LMA, i gel) is used to ventilate during CPR:

- Ventilate continuously and asynchronously with compressions
- DO NOT Pause compressions during ventilation
- The i-gel may be used as the initial, primary or exclusive airway

NOTE: **The mechanical ventilator may be used during CPR**

Ventilation Rate

- Adult—10 breaths/min
- Child/Infant—20-30 breaths/min accounting for age and clinical condition

⚠ CAUTION **Ventilation rates exceeding these recommendations may compromise hemodynamics.**

Defibrillation

Apply the PHI monitor/defibrillator as soon as practicable

- Identify the cardiac rhythm
- Defibrillate VF/VT as soon as possible after identification
- Defibrillate sustained or recurrent VT / VT q 2 min PRN after the first shock

Zoll X Series energy settings

Age Range	1st shock	2nd shock	3rd and subsequent
Adult	120J	150J	200J
Child / Infant	2J/kg—max 120J	4J/kg—max 150J	6J/kg—max 200J

Ongoing BLS

Perform ongoing BLS in coordinated 2-minute cycles with ≤ 10 sec pauses to perform:

- Compressor rotation
- Rhythm analysis
- Pulse check PRN
- Defibrillation PRN
- Monitor BLS and ensure optimal effective BLS is being used continuously
- Compressions should not be interrupted for advance procedures
- Perform BLS until return of spontaneous circulation (ROSC) or the resuscitation is terminated

4.1.2 ALS Guidelines

Capnography

Capnography provides physiologic information on ventilation, perfusion, and metabolism. If a patient has an advanced airway, capnography must be used to verify airway position, monitor ventilation, determine resuscitation effectiveness and idenify ROSC.

IV Access

Establish vascular access and administer an empiric Normal Saline (NS) fluid bolus PRN

- 20 cc/kg up to 500 mL
- Repeat PRN

Cardiac Arrest Refractory to BLS

Epinephrine—Cardiac Arrest

Adult

- 1 mg IV over 5-10 seconds
- 0.1 mg/mL concentration
- Repeat every 3–5 min until ROSC or resuscitation termination

Pediatric

- 0.01 mg/kg IBW IV over 5-10 seconds
- 0.1 mg/mL concentration
- Max single dose 1 mg
- Repeat every 3-5 min PRN

NOTE: **Administer as soon as feasible for non-shockable rhythms**

Sustained VF/VT Cardiac Arrest

An antiarrhythmic should be given for VF / VT cardiac arrest refractory to high quality BLS, defibrillation, and EPINEPHrine

Select one of the following:

Amiodarone

Adult

- 300 mg IV push
- Repeat 150 mg IV push PRN 1 time

Pediatric

- 5 mg/kg IV push
- Maximum single dose 300 mg
- Repeat 5 mg/kg PRN up to 2 times
- Maximum total dose 15 mg/kg

Lidocaine 2%—VF/Pulseless VT

Adult

- 1-1.5 mg/kg IV
- May repeat 0.5–0.75 mg/kg IVP every 5-10 min PRN
- Maximum total dose 3 mg/kg
- 2-4 mg/min maintenance infusion post ROSC

Ped

- 1 mg/kg IV
- May repeat 1 mg/kg after 5-10 min x 1 PRN
- Maximum single dose 100 mg
- 20-50 mcg/kg/min maintenance infusion post ROSC

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4.1.3 Reversible Causes

Once standard BLS and ALS treatment has been established review the events leading up to the arrest and perform a focused assessment.

- Systematically rule out or treat reversible causes based on the assessment findings.

AHA Reversible Causes of Cardiac Arrest (Hs and Ts)	
Hypovolemia	Tension pneumothorax
Hypoxia	Tamponade, cardiac
Hydrogen ion (acidosis)	Toxins (overdose)
Hypo-hyperkalemia	Thrombosis, pulmonary
Hypothermia	Thrombosis, coronary
NOTE: **Refer to applicable protocol, standing orders or contact medical control**	

NOTE: **Ultrasound may be used by qualified clinicians to assess myocardial contractility and identify potentially treatable causes of cardiac arrest**

- Advance airways must be monitored by capnography
- ETT placement and position must be confirmed regularly

⚠ CAUTION **Consider airway mal-position, obstruction, and ventilation device failure as the cause of any arrest in patients with advanced airways.**

Cross reference		
Clinical Care Guidelines	23	4-17 - Withholding and Terminating Resuscitation

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4.1.4 Special Circumstances

Clinical circumstances may require specialized management to ensure optimal patient care. Examples include when the primary etiology of arrest is respiratory, trauma, or electrolyte abnormalities. Based on the assessment findings, consider and apply additional treatment for any following special circumstances.

Trauma

- Reassess airway patency, ventilation, and oxygen delivery

NOTICE **Where practicable, recheck the ETT by directly observing it passing into the glottic inlet using video laryngoscopy.**

- Control all external bleeding with pressure, hemostatic dressings, and tourniquets
- Align and expediently splint long bone fractures
- Apply a pelvic compression device
- Administer blood products if available
- Empiric bilateral thoracostomy may be performed with evidence of chest trauma
- Pericardiocentesis may be performed with ultrasound evidence of pericardial effusion, or empirically with penetrating chest trauma

Polymorphic VT, Torsade de Point, or Prolonged QT

Magnesium

Adult

- 2 gm IVP

Pediatric

- 25–50 mg/kg IVP
- Max dose 2 gm

Pregnancy

- Displace the uterus manually to the left if the fundus is at or above the umbilicus
- Vascular access should be established above the diaphragm

NOTE: **If practicable, transfer to a clinician capable of peri-mortum C-section if refractory to treatment for 5 min**

Accidental Hypothermia

- Prevent further heat loss
- Active rewarming as soon as practicable
- Establish an advanced airway
- Give vasopressors and defibrillate as per standard guidelines

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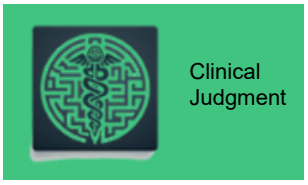
4.1.5 New Content



When managing a trauma patient who has lost pulses, chest compressions should be withheld unless there is a suspected medical cause or the presence of a shockable dysrhythmia. Prioritize lifesaving interventions such as blood transfusions over chest decompression, as compressions may interfere with these critical procedures.

Hyperkalemia

- Lab confirmed hyperkalemia—potassium >5.5 mEq/L
- History consistent with hyperkalemia—severe kidney dysfunction, dialysis, rhabdomyolysis, crush injury

NOTE: **Give either calcium gluconate or calcium chloride**

Calcium Gluconate

Adult

- 3 gm IV over 2–10 minutes
- Repeat every 10 minutes PRN up to 3 times

Pediatric

- 60 mg/kg IV over 2–10 minutes
- Repeat in 10 minutes PRN
- Maximum single dose 3 gm

NOTE: **Faster rate of IV administration may be indicated in cardiac arrest, cardiovascular instability and crush injures.**

NOTE: **Flush the IV line after administration**

Calcium Chloride

Adult

- 1 gm IV over 2–10 min
- Repeat every 10 minutes PRN up to 3 times

Pediatric

- 20 mg/kg IV over 2–10 min
- Repeat in 10 minutes PRN
- Maximum single dose 1 gram

NOTE: **Faster rate of IV administration may be indicated in cardiac arrest, cardiovascular instability and crush injuries**

NOTE: **Flush the IV line after administration**

Calcium Advisories

- Mixing calcium with other medications may cause a precipitate to form in the IV line
- Administer to neonates with on-line medical direction
- Rapid administration may cause bradycardia and hypotension
- Monitor patients taking digoxin for acute toxicity which can develop with calcium infusion

Sodium Bicarbonate

Adult

- 50 mEq IV over 5–10 min

Pediatric

- 1 mEq/kg IBW IV over 5–10 min
- Max single dose 50 mEq

NOTE: **May give for hyperkalemia when there is suspected or lab documented metabolic acidosis**

NOTE: **May glve for cardiac arrest due to hyperkalemia, bicarbonate is NOT indicated in cardiac arrest otherwise**

NOTE: **Give IV push in cardiac arrest**

NOTE: **May give 1 dose—additional doses require a physician order**

4.1.6 Post Arrest Management

Systematically assess and manage patients with a return of spontaneous circulation (ROSC) to ensure oxygenation and perfusion is optimized, identify the need for coronary intervention, and initiate temperature management is if indicated.

Ventilation and Oxygenation

- Titrate FiO2 to SpO2 > 93-99%
- Avoid excessive ventilation
- Titrate ventilation to EtCO2 35-40 mm/Hg

Perfusion

- Maintain SBP > 90 or 5th percentile for age
- Administer an empiric 1 to 2 L IV bolus in adults
- Administer vasopressors as per Shock & Cardiovascular Instability

12 Lead

- Perform a 12 lead ECG as soon a practicable
- Initiate care as per Acute Coronary Syndrome PRN

Post Arrest Temperature Management

Adult patients who have experienced a non-traumatic cardiac arrest and have a return of spontaneous circulation (ROSC) and meet the following criteria should be evaluated for cooling and targeted temperature management.

- Comatose
- Mean arterial pressure (MAP) sustained at 65 mm Hg or higher
- Less than 6 hours of return of spontaneous circulation

NOTICE **If there is a doubt about initiating therapy, crewmembers should contact medical control.**

Procedure

- Upon arrival, verify eligibility of patient for therapeutic hypothermia
- Establish invasive continuous core temperature monitoring

If cooling is required

- Place cold packs in axilla, on the lateral neck, and groin
- Administer chilled normal saline until target temperature has been reached.
- Target patient core temperature – 32C to 36C consistent with local practice

4.2 Bradycardia

Determine if the patient is symptomatic (altered mental status, hypotension, chest pain) and if the bradycardia is causing the symptoms. Assess for and treat underlying causes then treat bradycardia as indicated in the guideline.

AHA Reversible Causes (Hs and Ts)	
Hypovolemia	Tension pneumothorax
Hypoxia	Tamponade, cardiac
Hydrogen ion (acidosis)	Toxins (overdose)
Hypo-hyperkalemia	Thrombosis, pulmonary
Hypothermia	Thrombosis, coronary
Refer to applicable guideline or procedure	

- Administer O2 at the highest FiO2 available
- Establish venous access
- Perform a 12 lead ECG PRN

4.2.1 Adult Bradycardia

Symptomatic bradycardia (HR < 50 bpm or physiologically inadequate rate

Atropine

Adult

- 1 mg IV over 30 sec
- Repeat every 3-5 min PRN
- Max total dose 3 mg

Normal Saline (NS) or Lactated Ringers (LR)

- 250 to 500 cc Fluid bolus
- Repeat NS or LR PRN

Any of the following may be used as an alternative or adjunct to atropine

Transcutaneous pacing

- Apply and use as per the manufactures instructions
- Begin at ~ 10 mA in demand mode and titrate up by 5-10 mA to capture, pacer rate matched palpable pulses, and clinical improvement
- Full mA may be used initially and then titrated down for markedly unstable patients.

Transvenous pacing (if in place)

- Set mode, rate, output, and sensitivity as ordered by sending physician or on line medical control
- Verify capture, pacer rate matched palpable pulses, and clinical improvement

Epinephrine Infusion

Dopamine Infusion

4.2.2 Pediatric Bradycardia

Symptomatic Bradycardia (Physiologically Inadequate Rate)

- Perform CPR if HR remains < 60 bpm with poor perfusion despite adequate oxygenation and ventilation
- Normal Saline (NS) fluid bolus 20 cc/kg repeat PRN

Epinephrine

Pediatric

- 0.01 mg/kg IBW IV over 5-10 seconds
- 0.1 mg/mL concentration
- Max single dose 1 mg
- Repeat every 3-5 min PRN

EPINEPHrine Pediatric Bradycardia Dose Card

EPINEPHrine Pediatric Dose Guide (0.1 mg/mL conc.)					
Handtevy Color					
Age (yr)	1	3	5	7	9
Weight (kg)	10	15	20	25	30
Dose (mg)	0.1	0.15	0.2	0.25	0.3
Volume (mL)	**1 mL**	**1.5 mL**	**2 mL**	**2.5 mL**	**3 mL**

Bradycardia Due to Increased Vagal Tone or Primary AV Block

Atropine—Symptomatic Bradycardia

Pediatric

- 0.02 mg/kg IV over 30 sec
- Repeat every 3-5 min PRN
- Max single dose 0.5 mg
- Max total dose 1 mg

NOTE: **For bradycardia due to increased vagal tone or AV block**

4.3 Tachycardia

On initial contact with the patient obtain a handover report, perform a focused cardiac assessment, assess for reversible cause(s) of the tachycardia and provide standard cardiac treatment.

4.3.1 Assessment and Treatment Guidelines

- Monitor cardiac rhythm, SpO2, EtCO2, and NIBP
- Ensure an adequate airway and establish an advance airway as indicated
- Optimize ventilation and administer O2 PRN titrated to a SpO2 of 95 to 99%
- Perform and evaluate a 12 lead ECG
- Ensure adequate venous access, determine the patients' ability to tolerate fluids, and then give IV fluids as per the Shock and CV instability Guideline before administering antiarrhythmic medications.

AHA Reversible Causes (Hs and Ts)	
• Hypovolemia	• Tension pneumothorax
• Hypoxia	• Tamponade, cardiac
• Hydrogen ion (acidosis)	• Toxins (overdose)
• Hypo-hyperkalemia	• Thrombosis, pulmonary
• Hypothermia	• Thrombosis, coronary
Refer to applicable guideline or procedure	

4.3.2 Determine Cardiovascular Stability

Determine if the patient is unstable or symptomatic

- Unstable refers to a condition like chest pain, shortness of breath, or hypotension in which organ function is impaired, cardiac arrest is imminent and immediate intervention is indicated.
- Symptomatic implies that an arrhythmia is causing symptoms, such as palpitations, lightheadedness, or dyspnea, but the patient is stable and not in imminent danger.

NOTE: **In both unstable and symptomatic cases the medical crew must confirm it is the arrhythmia that is causing the patient to be unstable or symptomatic. Hypotension, hypoxia, and other clinical reasons must be ruled out prior to treating the arrhythmia.**

4.3.3 Identify the Rhythm and Categorize the Tachycardia

Determine the rhythm, rate, regularity, and QRS width to confirm the tachycardia is a *tachyarrhythmia*

- Adult Narrow (< or = 0.12 sec) or wide (> 0.12 sec)
- Pediatric Narrow (≤ 0.09 sec) or wide (> 0.09 sec)
- Regular or irregular
- Determine the rhythm
- Determine rate (Tachyarrhythmias are typically - Adult > 150, Child > 180, Infant > 220 BPM)

Unstable Tachyarrhythmias

NOTE: **Defibrillate as per the Cardiac Arrest Guideline if the rhythm is wide and irregular**
NOTE: **A Fib/A Flutter initial energy selection should be 150-200 J**

1. Procedural Sedation PRN
2. Antiarrhythmic medications may be utilized while cardioversion is being prepared
3. Perform cardioversion

Cardioversion Settings for ProPaq MD			
Age Range	**1st shock**	**2nd shock**	**3rd & subsequent**
Adult	100J	150J	200J
Pediatric	1J / KG	2J / KG	2J / KG

NOTE: **Adult tachyarrhythmias are typically > 150 BPM**

Adult Regular Wide Complex Tachyarrhythmias

Wide complex (QRS > 120 milliseconds) tachyarrhythmia

Amiodarone

Adult

- 150 mg mixed in 100 mL IV over 10 minutes
- Repeat 150 mg every 10 min PRN

Amiodarone—Maintenance Infusion

Adult

- 1 mg/min for 6 hours, then
- 0.5 mg/min for 18 hours

Lidocaine 2%

Adult

- 1-1.5 mg/kg IVP
- May repeat 0.5–0.75 mg/kg IVP every 5-10 min PRN
- Maximum total dose 3 mg/kg
- 2-4 mg/min maintenance infusion

Adult Irregular Wide Complex Tachyarrhythmias

Irregular wide complex (> 120 milliseconds) tachyarrhythmias. Determine the rhythm and treat accordingly

Polymorphic Ventricular Tachycardia Associated with QT prolongation

Magnesium—Polymorphic VT/QT Prolongation

Adult

- 1–2 gm IV over 15–20 minutes

Atrial Fibrillation | Wolfe Parkinson White

Atrial fibrillation pre-excited tachycardia associated with accessory pathways/Wolfe Parkinson White syndrome

Procainamide—Tachyarrhythmias

Adult

- Loading dose: 20 to 50 mg/minute until a maximum of 17 mg/kg IBW is administered, arrhythmia is controlled, or hypotension occurs
- Follow successful suppression of the arrhythmia with a 1–4 mg/minute continuous Infusion

WARNING **Adenosine should not be given for unstable, irregular, or polymorphic wide-complex tachycardias, as it may cause rhythm degeneration to VF**

Adult Regular Narrow Complex Tachyarrhythmias

Adenosine

Adult

- 12 mg rapid bolus over 1-3 sec
- Repeat 12 mg over 1-3 sec if no response in 1-2 minutes

Esmolol

Adult

- Initial load: 500 mcg/kg IV over 1 min
- Titrate 50 mcg/kg/min every 5 min to target HR or BP
- Repeat the loading dose of 500 mcg/kg before every titration PRN until near target HR
- Max infusion dose 300 mcg/kg/min

Pediatric

- NA

NOTE: **Give adenosine first, then Esmolol if adenosine response is inadequate**

Irregular Narrow Complex Tachyarrhythmias

Establish ventricular rate control for narrow complex atrial fibrillation or atrial flutter

Diltiazem

Adult

- 0.25 mg/kg TBW IV over 2 minutes
- Reassess after 15 minutes
- Repeat 0.35 mg/kg TBW IV if response inadequate
- Follow with a maintenance IV infusion at 5 to 15 mg per hr
- Titrate infusion to target heart rate

NOTE: **Control of rapid ventricular rates in atrial fibrillation or atrial flutter**

Pediatric

- NA

Pediatric Wide Complex Tachycardia

Amiodarone

Pediatric

- 5 mg/kg IV over 20 – 60 min
- Maximum single dose 300 mg,
- Repeat 5 mg/kg IV over 20–60 min PRN up to 2 times

Procainamide

Pediatric

15 mg/kg IBW IV/IO over 30–60 minutes

Pediatric Narrow Complex Tachycardia

NOTE: **Child tachyarrhythmias are typically > 180 BPM**
NOTE: **Infant tachyarrhythmias are typically > 220 BPM**

Adenosine

Pediatric

- 0.1 mg/kg rapid bolus over 1-3 sec
- Repeat 0.2 mg/kg over 1-3 sec if no response in 1-2 minutes
- Maximum single dose 12 mg

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4.4 Withholding and Terminating Resuscitation

Evaluate the futility of resuscitative efforts during the initial assessment of patients in cardiac arrest

- Exhaust all applicable and reasonable resuscitative efforts prior to using criteria in this guideline to terminate resuscitation
- Stratify patients into Adult, Pediatric or Neonatal age groups and then use age specific criteria
- Consult with medical control prior to terminating resuscitation if it is locally required, preferred, or uncertainty in meeting criteria exists.

4.4.1 Prolonged or In Route Resuscitation

The following clinical and operational situations that may support prolonged or in route resuscitation prior to terminating efforts.

- Local requirement or expectations
- Prolonged resuscitation is a bridge to definitive lifesaving treatment such as ECMO, c-section, therapeutic hypothermia, or re-warming.
- Communication with or access to the family is necessary
- The risk of termination to providers, family, or bystanders is greater than continuing resuscitation (e.g. risk of assault on scene)

4.4.2 Withholding Resuscitation

Medical Crewmembers may determining death and withhold or discontinue resuscitation if one or more of the following criteria exist:

- Resuscitation poses a risk of serious injury or death to the medical crew or other responders
- Clinical signs of irreversible death are present such as rigor mortis, dependent lividity, decapitation, incineration, decomposition, or massive trauma to the brain or heart.

NOTE: **During multiple-casualty incidents abbreviated assessment parameters such persistent apnea after repositioning of the airway may be used to determined death.**

Advanced Directives and DNR Orders

Medical crewmembers should assess for and abide by portable medical orders, do not resuscitate (DNR) orders, and pre-hospital care directives.

- Adhere to the various laws and regulations regarding advance care directives in the state the patient is in.
- To the degree possible, ensure advance directives are valid and properly executed.
- The limits of care dictated in advanced directives should be translated into medical orders by physicians or other qualified providers.
- If practicable, the relevant details of DNR orders and pre-hospital care directives should be discussed and confirmed with the patient, family, or attending physician.
- Any concerns or discrepancies should be reconciled with the transferring physician or medical control.
- If practicable, determine a plan for in route death prior to departing the patient rendezvous point,
- Clearly detail and document DNR orders, the content of conversations, decisions made, and rationale in the EHR.

NOTICE **The patient should receive all usual resuscitative care if a pre-hospital advanced care directive cannot be validated.**

Withholding Newborn Resuscitation

Medical Crewmembers may determine death and withhold or discontinue newborn resuscitation if one or more of the following criteria exist:

- Confirmed gestational age less than 23 weeks (Contemplate reliability of gestational age assignment)
- Birth weight less than 400 g
- Confirmed anencephaly
- Confirmed lethal genetic disorders or malformation

NOTE: **Exceptions to withholding resuscitation may be appropriate to fulfill parental wishes**
NOTE: **Medical Control should be contacted prior to discontinuing newborn resuscitation**

4.4.3 Termination of Resuscitation

Medical Crews may terminate resuscitation if all the following clinical and age criteria are met.

- Comprehensive resuscitative efforts that include high quality, minimally interrupted chest compressions, advance life support, and treatment for reversible causes have been accomplished.
- Apnea
- Absence of a central pulse
- The predominate rhythm is asystole or PEA less than 40 beats per minute
- Sustained EtCO2 < 20 mm/Hg via an advanced airway deemed functional and properly positioned by the Medical Crew

Additional Adult Specific Criteria

- No sustained return of spontaneous circulation (ROSC) after **at least 15 minutes** of resuscitative efforts.
- If required, consultation with medical control

Additional Pediatric Specific Criteria

- No sustained ROSC after **at least 30 minutes** of resuscitative efforts
- If required, consultation with medical control
- Transport is unsafe, impractical, or exceeds 30 minutes
- If practicable, the parents should be present and involved in the decision

Additional Neonatal Specific Criteria

- No detectable heart rate for **at least 10 minutes** of resuscitative efforts.
- If required, consultation with medical control or attending physician
- if practicable, the parents should be present and involved in the decision

Prolonged or In Route Resuscitation

The following clinical and operational situations that may support prolonged or in route resuscitation prior to terminating efforts.

- Local requirement or expectations
- Bridge to definitive lifesaving treatment such as ECMO, c-section, therapeutic hypothermia, or re-warming
- Communication with or access to the family is necessary
- Risk to providers, family, or bystanders is greater than continuing resuscitation (e.g. assault on scene)

4.4.4 Body Disposition and Additional Considerations

- Follow local policies regarding the disposition of the body after discontinuing resuscitative
- Provide emotional support to affected family members
- Maintain dignity of the victim

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4.5 Shock & Cardiovascular Instability

Shock is caused by inadequate delivery of oxygen relative to tissue demands. The following are signs and symptoms of shock.

- Anxiousness, confusion, or lethargy
- Cool, diaphoretic, pale, cyanotic, or mottled skin
- Delayed capillary refill
- Absence of distal pulses, unreadable distal SpO2, and poor distal pulse oximetry pleth
- Abnormal vital signs including tachypnea, tachycardia, hypocarbia, hypotension, shock index ≥ 1, and reduced urine output
- Lactate greater than ≥ 4 mmol/L
- Metabolic acidosis (pH < 7.35 with a HCO3 < 22)

NOTE: **Shock should be the 1st consideration for any patient with metabolic acidosis.**

4.5.1 Initial Contact

- Obtain a report.
- Perform a focused assessment.

4.5.2 Monitoring and Assessment

- Monitor heart rate, rhythm, respiratory rate, SpO2, EtCO2, and NIBP.
- Assess age-appropriate vital signs for changes consistent with shock and hypoperfusion.
- Document and assess vital signs every 5 minutes.

4.5.3 Identify and Treat Underlying Conditions

- Ensure or establish an adequate airway.
- Optimize ventilation and oxygenation.
- Identify and treat the specific underlying condition.

4.5.4 Improve Tissue Oxygenation

- Improve oxygen delivery to the tissues by transfusion.
- Establish venous access with two large bore IV or IO catheters.

4.5.5 Diagnostic and Interventional Measures

- Obtain a 12 lead ECG.
- Evaluate for STEMI or Non-STEMI.
- Activate interventional capabilities as indicated.

4.5.6 Temperature Control

- Control patient temperature with warm IV fluids, cabin heat, and blankets.

4.5.7 Medication Management

- Decrease analgesic, anesthetic, and anxiolytic medication doses to 1/2 of the normal amount based on patient response and clinical judgment.

4.5.8 Pediatric Vital Signs

Pediatric Vital Signs				
Color	**Age**	**Systolic Blood Pressure**	**Heart Rate**	**Resp Rate**
Gray	Neonate	60–100	100–160	30–60
Pink	4 mo	70–100	105–160	30–60
Red	6 mo	70–100	110–160	24–38
Purple	1 yr	75–105	90–150	22–30
Yellow	2 yr	75–110	100–160	22-30
White	3 yr	76-115	85-140	22-30
White	4 yr	78-115	75-120	22-26
Blue	5 yr	80-115	70-115	20-24
Blue	6 yr	82-120	70-115	20-24
Orange	7 yr	84-120	70-110	16-22
Orange	8 yr	86-120	70-110	16-22
Green	9 yr	88-120	65-105	16-22
Green	10 yr	90-120	60-100	16-22
Green	11 yr	90-120	60-100	16-22
Green	12 yr	90-120	60-100	16-22
Green	13 yr	90-120	60-100	16-22

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Hemodynamic Targets and Goals

Treatment should restore circulation to ensure minimally adequate organ perfusion. It may not be necessary to normalize all vital signs and it could be counterproductive to attain normal systolic blood pressures.

- Patients with head injuries, indications of increased intracranial pressure, spinal injuries or post cardiac arrest should be resuscitated to maintain a MAP of 80-100 mmHg.
- Other patients should be resuscitated to just maintain adequate tissue perfusion as evidenced by skin color, skin temperature, capillary refill, mentation, urine output, presence of radial pulses, or minimally acceptable BP.

NOTE: **Hemodynamic targets may be further defined for specific conditions in other guidelines.**

Initial Treatment and Stabilization

Clinical management and preparation for transport should be efficient and prioritized. Preparation time should be minimized if emergent surgical control or a time dependent intervention is necessary to manage the underlying disease such as:

- Interventional procedures—PCI, embolization, embolectomy
- Surgical Procedures—CABG, Vascular surgery, surgical hemorrhage control
- Treatments or devices—Balloon pump, VAD, ECMO

Intravenous Fluid Therapy

Empiric fluid administration may be use to treat undifferentiated shock and hypotension.

- Administer boluses of Lactated Ringers (LR) or Normal Saline (NS)
  - Adult—250-500 mL of LR or NS fluid boluses
  - Pediatric—10 to 20 mL/kg to a total of approximately 60 mL/kg
  - Newborn Infant—10 mL/kg to a total of approximately 30 mL/kg
- Evaluate patient response after each bolus and repeat PRN to achieve hemodynamic goals.
- Small precise boluses of crystalloid IV fluids should be administered in a systematic bolus—assess—repeat bolus fashion until the therapeutic goal is archived.

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4.6 Obstructive Shock

Early assessment and treatment should systematically identify, rule out, and manage the following causes of obstructive shock and instability:

Obstructive Shock Causes and Treatment		
Cause	Potential signs / symptoms	Treatment
Tension Pneumothorax	• Positive pressure ventilation JVD, Tracheal deviation • unilateral decreased or absent breath sounds	• Needle decompression and or • Simple thoracostomy
Cardiac Tamponade	• JVD, Narrowing pulse pressure, muffled heart tones • Alternating QRS axis on ECG • Ultrasound evidence of pericardial effusion	• Pericardiocentesis
Auto PEEP	• Positive pressure ventilation • High PIP + High P Plat + Auto PEEP on Vent • Shark fin capnogram	• Disconnect vent circuit • Allow pt to fully exhale • Slow rate and extend expiratory time
Acute Massive Pulmonary Embolism	• Chest pain, SOB, hypoxia refractory to oxygen, unilateral leg swelling, Sinus Tachycardia, Right Bundle Branch Block, S1Q3T3 / RV strain / T-wave inversion in V1-V3 + III on 12 lead ECG, HX of DVT, cancer, surgery or fracture	• Thrombolysis (on line medical direction) • Embolectomy

4.7 Cardiogenic Shock

Rule out or manage myocardial dysfunction by treating the causes:

- Dysrhythmias —Bradycardia, SVT, VT
- Electrolyte abnormalities — hypokalemia, hyperkalemia, hypomagnesemia**,** and hypocalcemia.

4.8 Septic Shock

Manage septic shock as per the Sepsis Guideline

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4.9 Hemorrhagic Shock

Control hemorrhage straight away using the techniques described in the trauma guideline.

- Apply direct pressure, pressure dressings, hemostatic dressings, and tourniquets to amendable wounds
- Apply pelvic compression devices for pelvic injuries
- Align and expediently splint long bone fractures

Tranexamic Acid—Uncontrolled Hemorrhage

Adult

- 2 gm IV over 1 min

Pediatric

- 25 mg/kg IV over 1 min
- Max dose 2 gm

NOTE: **May be given over a longer time consistent with local practices**
NOTE: **Use adult dosing for 16 and 17-year-old**
NOTE: **Use for significant hemorrhage with a SBP < than 90 mm/Hg or heart rate greater than 110 BPM**
NOTE: **Administer within 3 hours of injury or bleeding onset**

TXA Advisories

NOTICE **Prehospital administration of TXA may not be acceptable in all trauma systems. Medical Crewmembers must be knowledgeable of their regional TXA preferences and operate accordingly.**

- Do not mix with blood for transfusion or infusion solutions containing penicillin
- May administer as an infusion—usual mix 2 gm/100 mL (20 mg/mL) in D5W
- Optimal storage temperature 25 °C
- Not indicated in GI hemorrhage

4.9.1 Intravenous Fluid Therapy

IV fluids may be initiated for hemorrhagic shock and hypotension. LR or NS may be used as the lone resuscitation fluid, as blood is being prepared, or used after blood products have been depleted. The therapeutic goals vary depending on the whether the hemorrhage is controlled or uncontrolled.

NOTICE **Small precise crystalloid IV fluids boluses should be administered in a systematic bolus — assess — repeat fashion until the therapeutic goal is archived.**

Therapeutic goals

- Uncontrolled hemorrhage—titrate to the presence of a radial pulse and normal mentation
- Controlled hemorrhage— titrate to minimally normalized vital signs

Administration

Administer boluses of Lactated Ringers (LR) or Normal Saline (NS)

- Adult—250-500 mL
- Pediatric—10 to 20 mL/kg
- Newborn Infant—10 mL/kg

4.9.2 Blood Transfusion

NOTICE **Transfuse whole blood, packed red blood cells (PRBC), or plasma to patients who meet the established criteria as soon as practicable. Delaying transfusion can exacerbate the patient's condition, potentially leading to worsened outcomes such as hypoxia, shock, or organ failure due to inadequate oxygenation and perfusion.**

Transfusion Criteria

The following guidelines should be followed when administering blood products

Pediatric Criteria

- Uncompensated hypotensive shock (SBP < 70 mmHg + 2x age)

Adult Criteria

- Uncompensated hypotensive shock (SBP < 90 mmHg), **OR**
- Shock Index ≥ 1 and any one of the following:
  - Clinical suspicion of uncontrolled bleeding
  - Hemorrhage resulting in altered mental status

Blood Product Administration

Preferentially, administer physician-ordered ABO and RH compatible blood products if available.

- For adults, administer 1 unit of blood product.
- For children weighing 40 kg or less, administer 10 mL per kg.

If both plasma and packed red blood cells (PRBCs) are being used, alternate their administration using a 1:1 ratio.

Uncrossmatched Blood

If a patient has not been typed and crossed, or if ABO and RH compatible units are not available, administer one of the following:

- Low titer type O RH positive whole blood (LTOWB)
- O negative can be utilized in any situation
- Type O uncrossmatched packed red blood cells (PRBCs)
- Type AB or Type A uncrossmatched plasma

NOTE: **If LTOWB is administered, additional plasma is not necessary.**

Transfusion Procedure

- Standard blood filtration IV tubing must be used.
- Blood products and normal saline may be administered via the same IV line.
- Blood products must be warmed during infusion using an approved IV/blood fluid warmer.

NOTICE **Pressure bags may cause the flow to exceed the warming capacity of the IV/blood fluid warmer. Determine the relative importance of rapid infusion versus avoidance of hypothermia and adjust flow rate accordingly.**

NOTICE **Titrate NS or LR to maintain radial pulses and minimally adequate organ perfusion if blood products are unavailable or depleted.**

Reassessment

- Reassess after each transfusion
- The transfusion may be slowed as vital signs improve and evidence of adequate perfusion return
- Repeat the transfusion after 5-10 min if there is no improvement

NOTICE **Resuscitation to normalize vital signs is acceptable when hemorrhage has been definitively controlled (e.g. limb injury following application of a tourniquet).**

Calcium After Blood Transfusion

If there is no improvement in patient condition after 2 units or 2 pediatric weight based doses of blood products are transfused, calcium should be administered.

Calcium Gluconate—Blood Transfusion

Adult

- 3 gm IV over 2–10 minutes
- May repeat each time 2 units of blood products are given

Pediatric

- 60 mg/kg IV over 2–10 minutes
- Repeat each time 20 mL/kg of blood products are given
- Maximum single dose 3 gm

NOTE: **Give if there is no improvement after blood administration**
NOTE: **Flush the IV line after administration**

Calcium Chloride—Blood Transfusion

Adult

- 1 gm IV over 2–10 min
- Repeat each time 2 units of blood products are given

Pediatric

- 20 mg/kg IV over 2–10 min
- Repeat each time 20 mL/kg of blood products are given
- Maximum single dose 1 gram

NOTE: **Give if there is no improvement after blood administration**
NOTE: **Flush the IV line after administration**

Transfusion Reactions

Clinicians must remain vigilant in identifying, managing, and documenting transfusion reactions, ensuring patient safety through efficient implementation of appropriate interventions.

Types of Transfusion Reactions

Mild Reactions

- **Febrile Reaction:** A temperature increase of 1 °C above baseline.
- **Allergic Reaction:** Symptoms may include itching, rash, or hives.

Severe Reactions

- **Anaphylaxis:** A severe, life-threatening allergic reaction requiring immediate intervention.
- **Acute Hemolysis:** Resulting from ABO incompatibility, this reaction can cause back pain, chills, fever, and dark urine.
- **Transfusion-Associated Circulatory Overload (TACO):** Symptoms include shortness of breath, hypertension, and fluid overload.
- **Transfusion-Associated Acute Lung Injury (TRALI):** Characterized by acute respiratory distress and hypoxia.

Delayed Reactions

- **Disease Transmission:** Transmission of infectious agents not detected during screening.
- **Delayed Hemolysis:** A slower onset reaction that may occur days after the transfusion.
- **Graft vs. Host Disease (GvHD):** A rare but serious condition where donor white blood cells attack the recipient's tissues.

Management and Reporting Procedures

- **Immediate Response:** For any signs of a transfusion reaction, stop the transfusion immediately, maintain intravenous access with saline, and monitor vital signs closely.
- **Assessment and Intervention:** Conduct a thorough assessment to determine the type of reaction. Implement appropriate interventions based on the severity and type of reaction.
- **Documentation:** Document all reactions, assessments, interventions, and outcomes in the patient's medical record and the Blood Variance Report.
- **Notification:** Notify the receiving facility physician and the transfusion service/blood bank. For severe or delayed reactions, further investigation may be necessary to prevent recurrence.

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Non-Hemorrhagic Shock

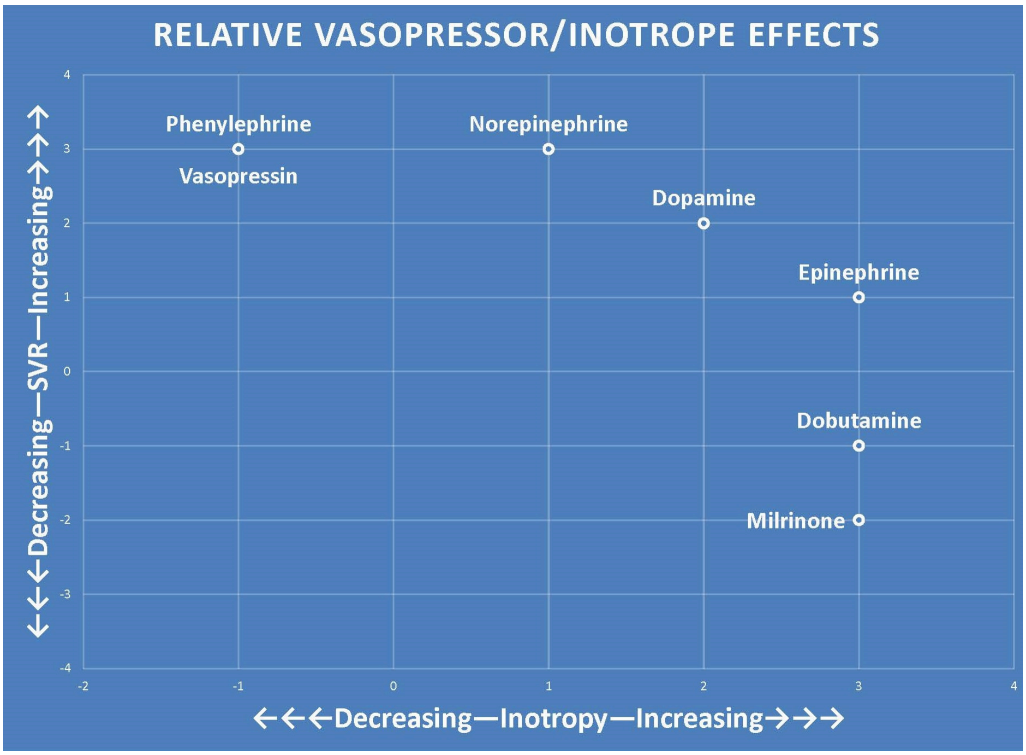
Non-hemorrhagic or non-obstructive shock that is refractory to fluid administration should be treated with a continuous IV vasoactive medication.

Vasopressors and Inotropes

- Vasopressors induce vasoconstriction
- Inotropes increase cardiac contractility
- Many drugs have both vasopressor and inotropic effects
- The choice of medication should be based on patient presentation, disease and concurrent treatment
- NORepinephrine is the vasoactive medication of choice
- EPINEPHrine is acceptable and may be preferred if the patient is bradycardic or a stronger Beta 1 agonist is needed
- Vasopressin may be administered as an additional second line agent for shock that is refractory to norepinephrine at doses greater than 10–20 mcg/min
- Vasopressors may be initiated concurrently with administration of IV fluids

Primary Vasoactive Medications

Medication	Adult Dosing	Ped Dosing	Receptor	Notes
EPINEPHrine	2–10 mcg/min	0.05–1 mcg/kg/min	Alpha 1 Beta 1	• î heart rate, î contractility, î vasoconstriction • Alternate Adult dosing 0.1-0.5 mcg/kg/min
Norepinephrine	2–35 mcg/min	0.05–2 mcg/kg/min	Alpha 1 Beta 1	• Vasoconstriction with small amount of beta 1 effects • Alternately in adults 0.05-2 mcg/kg/min
Vasopressin	0.04 units/min	NA for shock	V1	• May be used as a second agent
Secondary Vasoactive Medications				
Medication	**Adult Dosing**	**Ped Dosing**	**Receptor**	**Notes**
DOPamine	5–20 mcg/kg/min	5–20 mcg/kg/min	Beta 1 Alpha 1	• Moderate doses mainly increase heart rate & contractility • High doses mainly cause isolated Alpha 1 vasoconstriction
DOBUTamine	5–20 mcg/kg/min	5–20 mcg/kg/min	Beta 1 Beta 2	• Increased cardiac output • Vasodilation may lower blood pressure
Phenylephrine	100–180 mcg/min	0.1–3 mcg/kg/min	Alpha 1	• Typical steady state maintenance 40-60 mcg/min • An alpha 1 drug that causes isolated vasoconstriction



Continuation of Infusions

Vasoactive medications in place at the time of patient rendezvous should be continued if they are effective and consistent with the guidelines or standard practice.

NOTICE **If therapy at the time of patient rendezvous is ineffective, alternative treatment options consistent with this guideline should be implemented or discussed with the sending provider, receiving physician, and or medical control.**

Non-standard Infusion Dosing Schemas

If an existing infusion is being administered using a dosing schema inconsistent with PHI guidelines, do one of the following:

- Continue therapy using the existing dosing schema
- Recalibrate to the PHI standard dosing schema while leaving the dose delivered unchanged

NOTE: **In either case clearly document changes and the dosing schema used in the medication comments section.**

CAUTION **If an appropriate and effective infusion is in place at the time of rendezvous the delivered dose of medication should be continued. Arbitrarily changing the administration rate and delivered dose to match a dose schema may harm the patient.**

IV Push Pressor Therapy

IV push EPINEPHrine or phenylephrine may be used to treat transient or profound hemodynamic instability while administering volume or preparing vasoactive infusions. Each drug must be chosen based on its properties and therapeutic goals.

- EPINEPHrine is a potent vasoconstrictor, inotrope, and chronotrope suitable for all shock states except for those caused by tachyarrhythmias.
- Phenylephrine is pure alpha agent and potent vasoconstrictor suitable for hypotension secondary to anesthetic administration and vasodilatory shock.

WARNING **IV push pressors must be prepared and administered using the standard dilute push dose concentration for each pressor. Failure to do so may result in medication errors and patient harm.**

Phenylephrine Push Pressor Administration

NOTICE **IV push pressor phenylephrine should be used as a temporary treatment while preparing an infusion. Administration should be limited to 3 total doses.**

Phenylephrine—Push Pressor

Adult

- 50–200 mcg IV
- 100 mcg/mL concentration
- Repeat every 2–5 min PRN
- Administer up to 3 doses total

Ped

- 5–20 mcg/kg IV
- Max single dose 200 mcg
- 100 mcg/mL concentration
- Repeat every 2–5 min PRN
- Administer up to 3 doses total

Phenylephrine Push Pressor Preparation

WARNING **IV push pressor phenylephrine must be prepared and administered in the standard 100 mcg/mL concentration. Failure to do so may result in medication errors and patient harm.**

Bag Dilution

1. Add 10 mg (1 ml via) phenylephrine to a 100 mL bag of NS—final concentration 100 mcg/mL
2. Label the bag
3. Using an empty syringe, draw the push dose phenylephrine to be administered from the bag
4. Label the syringe

NOTE: **The 100 mcg/mL phenylephrine IV bag is also an acceptable infusion mix and may be used for a follow-on infusion**

Epinephrine Push Pressor Administration

IV push pressor EPINEPHrine should be used as a short term or temporary treatment while preparing an infusion. Administration should be limited to 3 total doses.

Epinephrine Push Pressor Preparation

IV Push Pressor EPINEPHrine must be prepared in one of the following ways and administered in the standard 10 mcg/mL concentration. Failure to do so may result in medication errors and patient harm.

Syringe Dilution

1. Transfer 0.1 mg (1 ml) from a 1 mg/10 mL (0.1 mg/mL) EPINEPHrine prefilled syringe to an empty 10 cc administration syringe
2. Add 9 mL of normal saline to the 10 mL administration syringe
3. Final mix 0.1 mg/10 mL (10 mcg/mL)
4. Label the syringe

NOTE: **Segregate the EPINEPHrine prefilled syringe from the push dose syringe by placing it in a pack or other area**

Bag Dilution

1. Add 1 mg (1 mL vial or ampule) EPINEPHrine to a 100 mL bag of normal saline
2. Final concentration 1 mg/100 mL (10 mcg/mL)
3. Label the bag
4. Using an empty syringe, draw the push dose EPINEPHrine to be administered from the bag
5. Label the administration syringe

NOTE: **The 1 mg/100 mL (10 mcg/mL) EPINEPHrine IV bag may be used for the follow-on infusion**

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Prefilled Flush Syringes

Commercial prefilled 0.9% sodium chloride syringes are regulated by the FDA as devices, not medications. These syringes have been approved for flushing vascular access devices, not the reconstitution, dilution or administration of IV medications.

Using prefilled 0.9% sodium chloride flush syringes for reconstitution, dilution or administration of medications is prohibited. The safety of using these devices for medication delivery has not been evaluated.

4.9.3 Special Circumstances

Catecholamine-resistant Shock

Evaluate for unrecognized conditions like tension pneumothorax, pericardial effusion, and ongoing blood loss.

Stress Dose Steroids

Consult Medical control if adrenal crisis is suspected and stress dose steroids are indicated.

Hydrocortisone

Adult

- 100 mg IV—usual dose
- Requires an online physician order

Pediatric

- No standing orders
- Requires an online physician order

III Infants and Congenital Heart Defects

III Infants may have a previously undiagnosed ductal dependent congenital heart defect resulting in rapid clinical deterioration refractory to fluids and vasopressors. Medical control should be consulted for alprostadil orders if a ductal-dependent congenital heart defect is suspected.

Alprostadil

- 0.02–0.05 mcg/kg/min

NOTE: **Requires on line order**

Pediatric Hypocalcemia

Contact medical control if persistent shock and symptoms of hypocalcemia or an ionized calcium <1.1 mmol/L (4.8 mg/dL) is encountered,

Calcium Gluconate

NOTE: **Requires on line order**

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